Critical Minerals And Rare Earths Supply Risk Analysis

Executive Summary

The critical minerals underpin renewable power generation, semiconductor manufacturing and multiple strategic sectors yet their supply is geographically narrow. Their supply chains are not always evenly distributed, mining and refining capacity can be concentrated in a small number of countries, while demand and prices can change quickly. This report analyses 24 minerals, 35 countries dataset covering 2015-2026 and track five business KPIs : Concentration Index, Top Country Dependence, Demand growth, Minerals under export control and Price Volatility. The underlying theme is that critical minerals issues are inherently supply chain issues. Establishing domestic mining and concentrate production does nothing to mitigate risks if downstream processing is highly concentrated geographical imported and unreliable It is important to note, in this context, that each of the minerals on the critical minerals list has its own supply chain and is in some way unique in terms of geologic occurrence, ore grade, host mineralogy, extractive metallurgy or other processing requirements, and hence each needs to be evaluated individually to develop commodity-specific mitigation strategies.

The analysis points out four important findings. First, refining is more concentrated than mining. This suggests that processing the minerals may have a greater concentration risk than extraction alone.
Second, country dependence remains substantial. In 2026, 17 minerals have top country dependence of at least 50%, while 9 exceed 70%, highest dependency is 91.7%. Third, demand growth differs by end use. Battery-related minerals show noticeably higher demand growth than other end-use categories. And Fourth, supply risk is not uniform across minerals or countries.

Taken together, the findings suggest that the organisations should look beyond the availability of the raw materials and pay equal attention to refining capacity, geographic concentration, export controls and price exposure.

Problem Statement

The growing dependence on critical minerals across industries such as renewable energy, electric vehicle manufacturing, electronics and advanced technologies has made their supply chains increasingly important to the global economy.

However, the availability and distribution of these minerals present several challenges, including geographical concentration of production, dependence on a limited number of supplier countries, fluctuations in market prices, increasing demand and government-imposed export restrictions. These challenges create uncertainty around the stability and reliability of critical mineral supply chains. In particular, heavy dependence on a small number of countries for mining and refining activities can increase the vulnerability of industries to geopolitical developments, trade restrictions and unexpected supply disruptions.

Through data analysis and interactive Power BI visualisations, the project seeks to support better understanding of supply-chain dependencies and identify areas where diversification, risk monitoring and improved resource planning may strengthen supply-chain resilience.

Introduction

The purpose of this project is to examine patterns in the global supply of selected critical minerals and to identify factors that may increase the risk of disruption. Rather than looking at the characteristics of each mineral as an individual commodity, we look at the entire supply chain: where the mining is done, how concentrated is the production/refining of these minerals, how is demand changing, is there any evidence of instability in prices.

Critical minerals and rare earths sit underneath almost every strategic technology like EV batteries, wind turbines, semiconductors, permanent magnets, and defense systems. A handful of countries control most of the world's mining and an even smaller handful controls refining, which makes the entire clean-energy and electronics supply chain sensitive to a small number of geopolitical decisions.

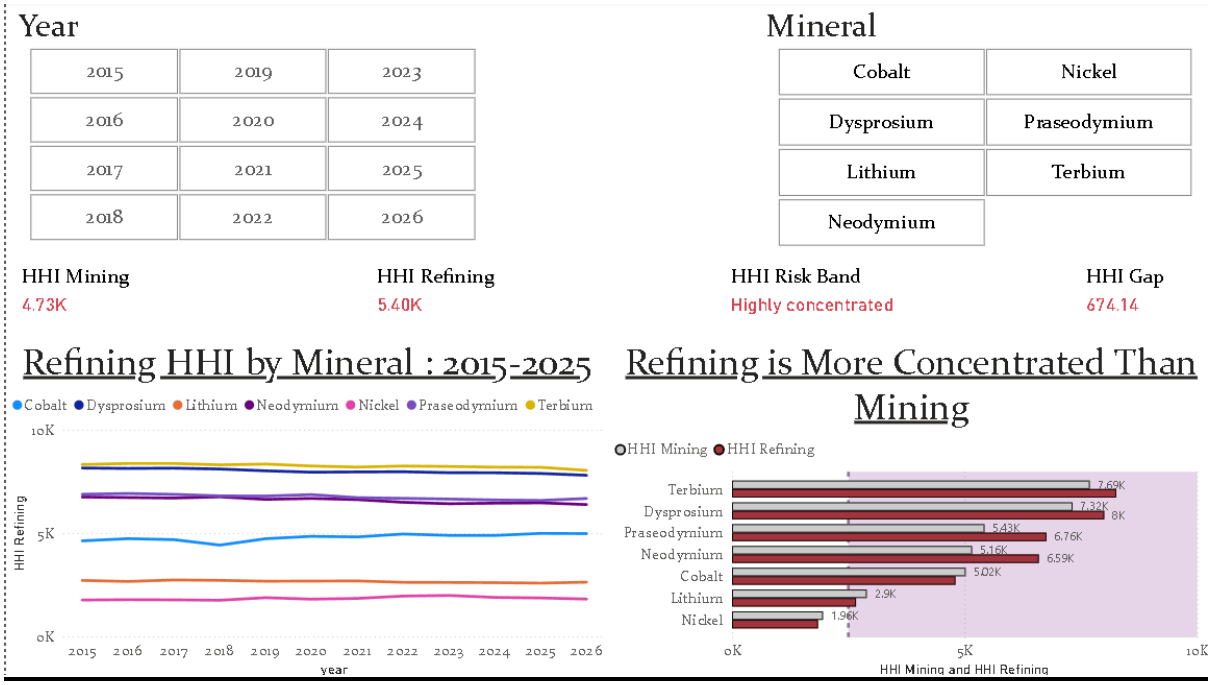
The dataset contains 1,548 records covering 24 minerals, 35 countries from the years 2015 to 2026. The fields include mine production, production share, reserves, years of reserves, refined share, price, demand growth, the Herfindahl–Hirschman Index (HHI), top-country share, supply-risk score, export-control status and disruption indicators. The dashboard presents these measures through trend charts, ranked comparisons, KPI cards and country-level views.

The following key performance indicators (KPIs) guide the analysis :

- HHI Concentration
- Top Country Dependence
- Demand Growth
- Minerals under Export Control
- Price Volatility

Analysis

Concentration Index (HHI)



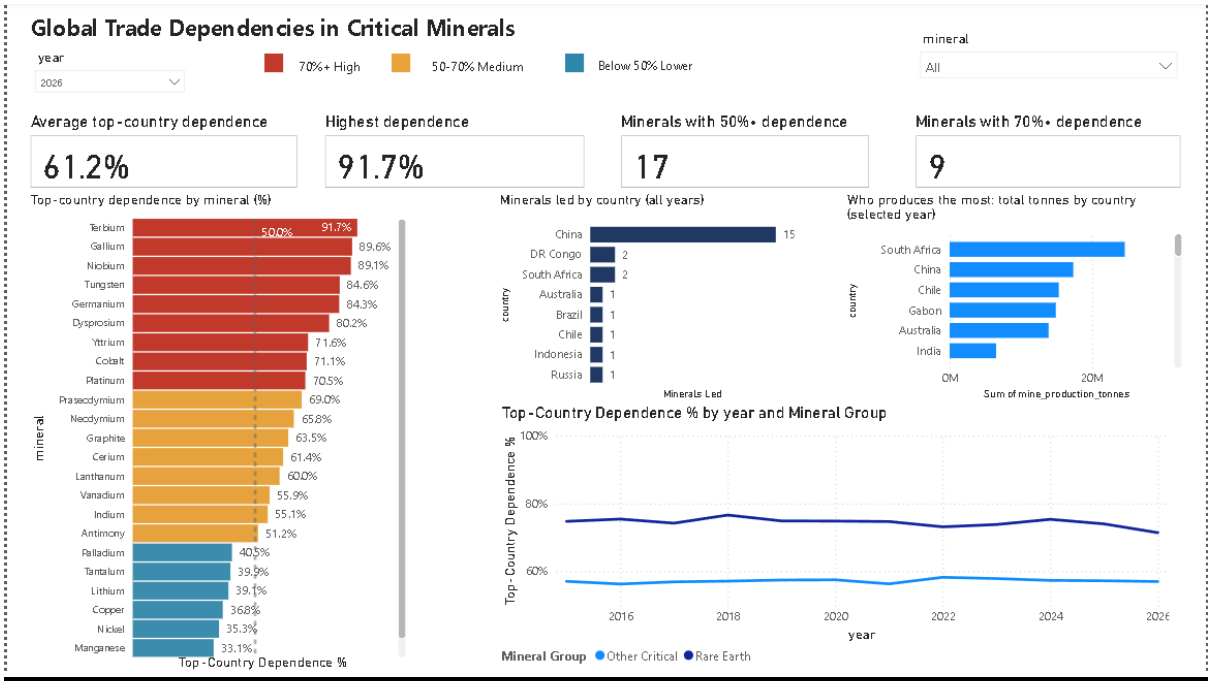
The Herfindahl-Hirschman Index (HHI) measures how concentrated supply is. Each country's share of supply is squared and the results are added together. Scores above 2,500 are classed as highly concentrated, indicating that a small number of countries hold enough market power to dictate supply conditions.

Overall result - Both stages are highly concentrated. Mining averages an HHI of 4.73K and refining 5.40K. Refining is therefore the more concentrated stage, although this average combines two quite different groups of minerals.

Rare earths - Refining is the bottleneck. Among the rare earths, refining is the much tighter stage. Neodymium goes from 5.16K in mining to 6.59K in refining and Praseodymium from 5.43K to 6.76K. Dysprosium and terbium are already high in mining and reach about 8K and 8.26K in refining respectively, which is near maximum for a market run by a single country, a likely explanation is that mining depends on the location of geological deposits. Refining needs capital, technical expertise and once one country has built that, their rivals find it hard to catch up.

Trend (2015-2025) - Refining HHI changed very little for any mineral over the period. Because new refining capacity takes years to build and concentration is unlikely to ease without deliberate practices. For the rare earths, this points to developing refining capacity outside the largest refiner.For cobalt, the priority is to diversify sourcing and hold strategic reserves.

Top Country Dependence



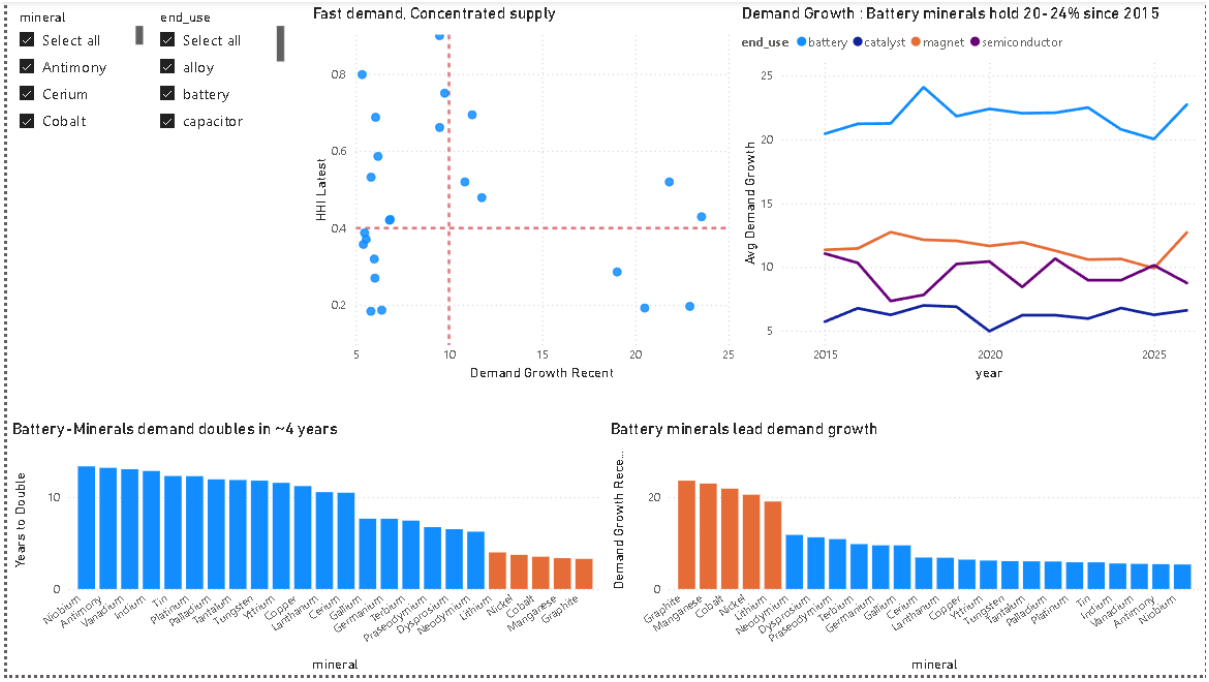
Top-country dependency measures the extent to which supply of a mineral is associated with its largest producing or refining country. It is a useful measure of geographic exposure, when a large share of supply depends on one country, buyers have fewer alternatives if exports are restricted or production is disrupted.

As shown above, 17 minerals have top country dependency above 50%, and nine are above 70%. The highest dependency is 91.7%. Terbium is shown at the top of the mineral-level comparison, followed by gallium and niobium. The underlying 2026 dataset also records very high average top-country shares for these minerals: approximately 91.7% for terbium, 89.6% for gallium and 89.1% for niobium.

Import dependence stays high throughout the entire period. Disaggregating the data by country and mineral reveals considerable variation that a single average conceals, and it indicates which minerals combine the highest exposure with the greatest difficulty of diversification. These minerals should therefore be prioritised in supply-security policy.

The ranked bar chart orders minerals by concentration, so those most dependent on a small number of suppliers appear at the top of the display. The trend line adds a dimension showing where the concentration is rising, falling or stable over the period of study. Filters for year, mineral and mineral group let users restrict the comparison to a specific subset of the data.

Demand Growth



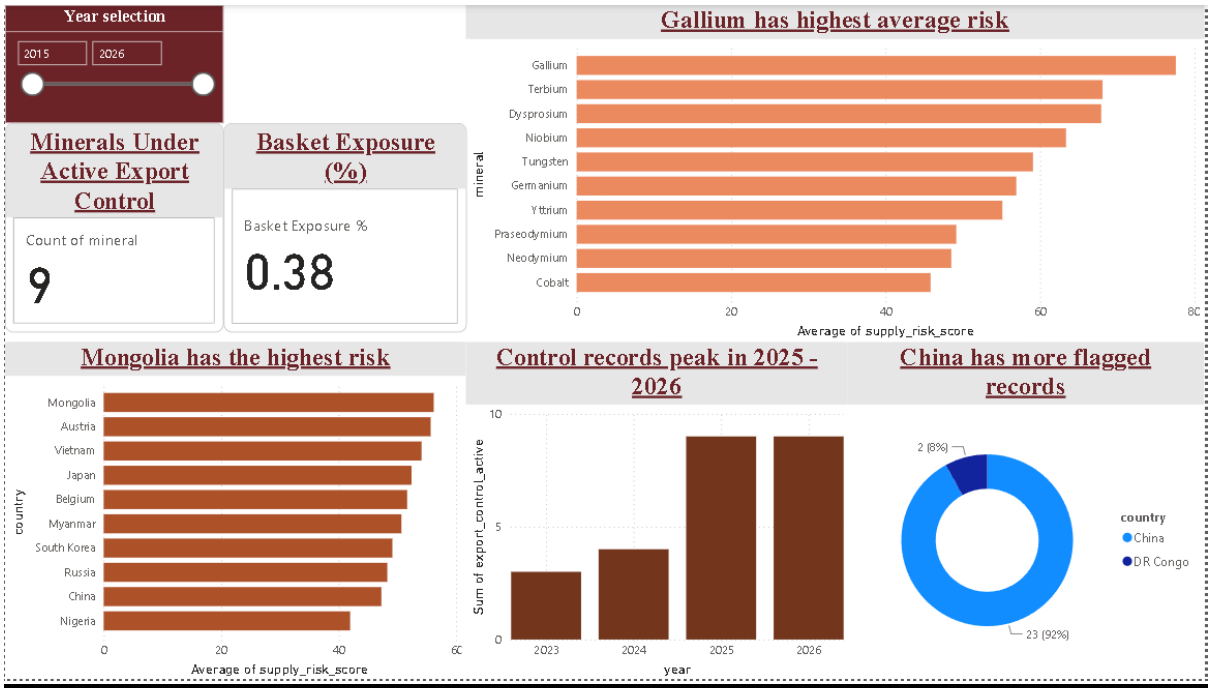
Demand growth matters for supply risk security because supply chains can come under strain when the consumption rises faster than production capacity can adjust. The dataset classifies minerals by end use across eleven categories, including battery, magnet, semiconductor and grid applications.

Battery minerals show the clearest pressure. Their average demand growth has stayed between about 20% and 24% since 2015, peaking near 24% in 2017 and standing at 22.7% in 2026. Magnet minerals average 12.7% in 2026, a rebound from about 10% in 2025, and semiconductor minerals average 8.7%, down from about 11% in 2015. Catalyst minerals remain between 5% and 7% throughout the period. At the mineral level, five minerals (graphite, manganese, cobalt, nickel and lithium) grow at roughly 19% to 24%. The next group grows at about 10% to 12%, so the gap is large. At the faster rates, demand for these five minerals doubles in about three to four years, compared with roughly six to thirteen years for the other minerals.

The scatter plot is used to show demand growth with supply concentration, the focus being the five minerals that fallin the upper-right quadrant. Two combine growth above 20% with HHI between roughly 0.4 and 0.5, and three combine growth of about 11% to 12% with HHI between roughly 0.5 and 0.7. The other three fast-growing minerals have HHI values of about 0.2 to 0.3, which suggests lower supply risk. Conversely, the most concentrated minerals (HHI

of about 0.75 or higher) grow at under 10%. Demand growth and concentration therefore do not coincide for most minerals, and the minerals that combine both deserve the closest monitoring.

Minerals under Export Control

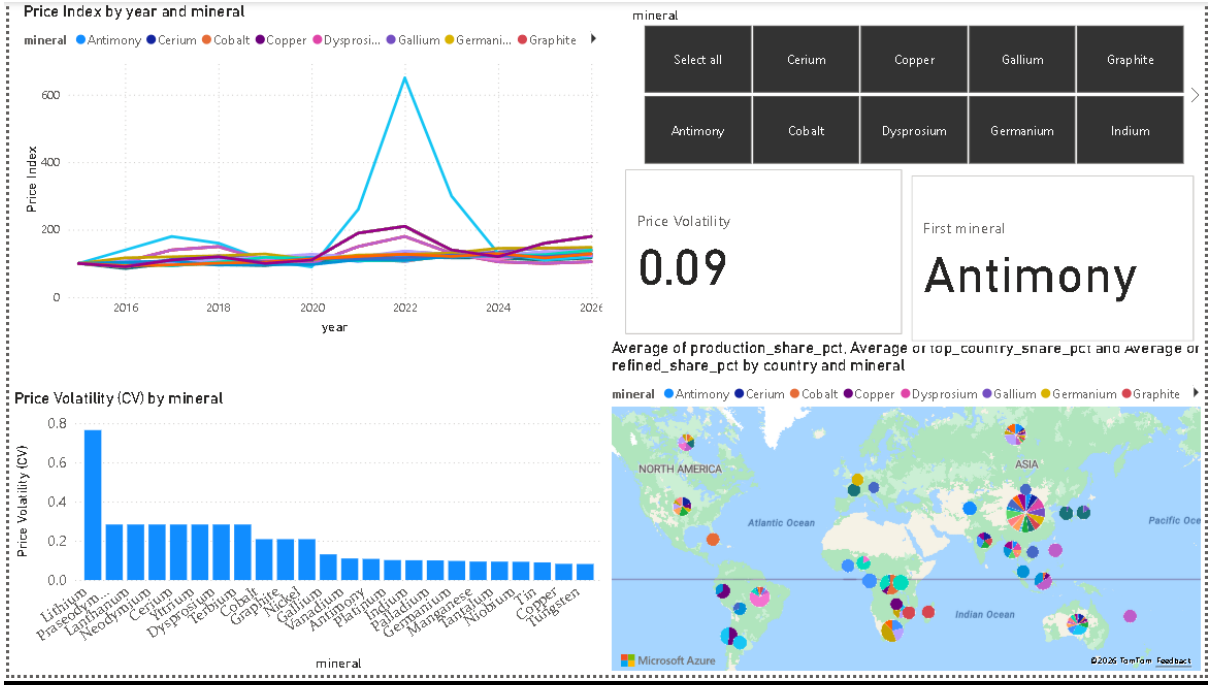


The export control dashboard summarises exposure to active controls through two indicators: the number of active controls, currently 9, and the share of the mineral basket exposed to those controls, currently 37.5%. Together they indicate that export controls are a significant risk for the basket. A year slicer allows the results to be compared across the 2015 to 2026 period.

Flagged records rose from 3 in 2023 to 4 in 2024 and 9 in 2025, then remained at 9 in 2026. The tripling between 2023 and 2025 indicates that control-flagged observations became considerably more common over the final years of the series, and the absence of further growth after 2025 suggests that the count has levelled off. The series counts binary flags across dataset rows, and a mineral can appear in several rows across years and countries. It therefore cannot show that new laws or restrictions took effect, nor that the number of distinct controlled minerals tripled. By country, China accounts for 23 flagged records and the Democratic Republic of Congo for 2, so control flags in this dataset are concentrated overwhelmingly in China-related records.

Mongolia has the highest average supply-risk score among countries, and gallium has the highest among minerals. Minerals with high risk scores should therefore be examined alongside their export control status. The risk scores function as a screening tool: they identify where supply risk may be concentrated and where alternative sources may be needed, but they do not show that a shortage, disruption or price spike has occurred. Further investigation is required before drawing conclusions about any individual mineral or country.

Price Volatility



Price volatility is measured here with the coefficient of variation (CV), calculated as the standard deviation divided by the mean price of each mineral. Because the CV is scale-free, it allows minerals with very different average prices to be compared directly. A high CV indicates unstable relative prices, and a low CV indicates stable ones.

Lithium is the most volatile mineral, with a CV of about 0.76, more than two and a half times that of any other mineral. The rare earth minerals (praseodymium, neodymium, dysprosium and cerium) are also volatile, at about 0.28. Cobalt and nickel are moderately volatile, at about 0.21, while copper and tungsten are relatively stable, at about 0.08 to 0.09. Lithium therefore carries the greatest relative price uncertainty of the minerals examined.

The price index series for 2015 to 2026 supports this ranking. Lithium's index rose by about 550% between 2015 and 2022, from a baseline of 100 to roughly 650, and then fell by about

83% from that peak to roughly 110 in 2025. By 2025, the indices for cerium, neodymium and dysprosium had each reached 160, a 60% increase over the baseline. The indices for cobalt and nickel stood at 100, indicating no net change, and copper's index was 118.99, about 19% above the baseline.

These comparisons show that the current price level does not reflect historical volatility. Lithium's 2025 index was below those of several rare earth minerals even though its CV is far higher, because the CV captures the size of price swings over the whole period, not the net change between two points.

Supplier diversification, long-term supply contracts and regular price monitoring would probably reduce exposure to unexpected price swings. Combining historical price data with volatility measures would also support better-informed procurement, inventory management and investment decisions.

Recommendations

Diversify refining capacity

Refining concentration is higher than mining concentration, so diversification should address processing as well as extraction. Organisations that depend on a few refining locations should assess alternative processors, arrange secondary supply and, where commercially viable, support new refining capacity.

Prioritise highly dependent minerals

Minerals for which a single country supplies more than 70% of demand should be the first candidates for supply-chain review. In 2026, terbium, gallium and niobium will exceed this threshold. Buyers should map their direct suppliers and identify alternative sources or substitute materials where these exist.

Strengthen export-control monitoring

Control-flagged records increased in the later years of the dataset, which supports regular monitoring of trade policy. Organisations should keep a current register of relevant restrictions and assess how a change in one country's export policy would affect their suppliers and production schedules.

Conclusion

Critical-mineral supply risk depends on more than the volume mined. Refining concentration, reliance on a few supplier countries, shifting demand, export controls and price volatility each affect how resilient a supply chain is. Dependency on a single country is high for several minerals, risk scores are highest for gallium and a group of rare-earth-related materials, and demand growth is strongest for battery minerals (22.7% in 2026). Organisations need to know where their materials are processed, how far they depend on individual countries, and which minerals combine rapid demand growth with concentrated supply. Regular monitoring and supplier diversification would reduce exposure and support better-informed procurement.

References

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Assessing the supply risk of geopolitics on critical minerals for energy storage technology in China
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